

PAPER_658 — LQG Black Hole Bounce with UQFF Vacuum Density Elevation

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Source: grok_share_fc21e30c24b4.txt — BlackHoleBounce class (May 2025)

Version: v5.28

UQFF Framework: Vacuum Density Series (PAPER_646) integration

C++ Module: BlackHoleBounceUQFF.h / BlackHoleBounceUQFF.cpp

CP4 Entry: #242

Abstract

Loop Quantum Gravity (LQG) cosmology replaces the Big Bang singularity with a “bounce” — a quantum-gravitational rebound at Planck-scale densities. The standard Loop Quantum Cosmology (LQC) Friedmann equation introduces a critical density ρ_c that prevents singularity formation. This paper extends LQC with the Unified Quantum Field Framework (UQFF), incorporating the Vacuum Density Series constants ρ_{UA} and ρ_{SCm} . The UQFF elevates the critical density by a factor of $(1 + \rho_{UA}/\rho_{SCm}) \approx 11$, extending the bounce energy scale and primordial black hole (PBH) lifetime by the same factor. A UQFF-modified scale factor, effective equation of state, and simulation protocol are derived and validated numerically.

1. Introduction

The Big Bang singularity problem is a fundamental tension in cosmology: General Relativity predicts the universe emerged from a zero-volume, infinite-density state, which is physically unacceptable. LQG resolves this by quantising spacetime geometry; in the LQC reduction, the universe “bounces” from a prior contracting phase at the Planck density ($\sim 5.16 \times 10^{96} \text{ kg/m}^3$).

The UQFF (Murphy, 2025-2026) posits that two vacuum density fields permeate spacetime: - **Universal Aether [UA]:** $\rho_{UA} = 7.09 \times 10^{-36} \text{ J/m}^3$ - **Superconductive Medium [SCm]:** $\rho_{SCm} = 7.09 \times 10^{-37} \text{ J/m}^3$

Their ratio ($\approx 10:1$) and the time-reversal factor $f_{TRZ} = 0.1$ appear throughout UQFF as negentropic modifiers. This paper introduces those modifiers into LQC.

2. Standard LQC Friedmann Equation

The LQC-modified Friedmann equation replaces the classical $H^2 = (8\pi G/3)\rho$ with:

$$H^2 = \frac{8\pi G}{3} \rho \left(1 - \frac{\rho}{\rho_c}\right) - \frac{kc^2}{a^2}$$

where: - $H = \dot{a}/a$ (Hubble parameter) - ρ = matter/energy density - ρ_c = critical bounce density $\approx 0.41 \rho_{Pl} \approx 5.16 \times 10^{96} \text{ kg/m}^3$ - k = spatial curvature ($k = 0$ for flat universe)
- a = scale factor

At $\rho \rightarrow \rho_c$, $H \rightarrow 0$: the expansion stalls. For $\rho > \rho_c$ the argument goes negative; in the quantum theory this corresponds to the forbidden bounce region. The scale factor near the bounce is:

$$a(t) \approx a_{\min} \cosh\left(\frac{t}{t_{\text{Pl}}}\right)$$

with Planck length $a_{\min} = \sqrt{(\hbar G/c^3)} \approx 1.62 \times 10^{-35}$ m and Planck time $t_{\text{Pl}} = \sqrt{(\hbar G/c^5)} \approx 5.39 \times 10^{-44}$ s.

3. UQFF Modification

3.1 Elevated Critical Density

The UQFF vacuum fields add a density-ratio correction to ρ_c :

$$\rho_{c,\text{UQFF}} = \rho_c \cdot \left(1 + \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}}\right)$$

Numerically:

$$\rho_{c,\text{UQFF}} = \rho_c \cdot \left(1 + \frac{7.09 \times 10^{-36}}{7.09 \times 10^{-37}}\right) = 11 \rho_c$$

Physical interpretation: The [UA] field provides an upward negentropic pressure that raises the energy barrier at which the bounce occurs. This extends the quantum bounce into a higher-energy regime, with direct implications for PBH formation rates and lifetimes.

3.2 UQFF Scale Factor

Incorporating both the f_{TRZ} time-reversal factor and the ρ -ratio buoyancy expansion:

$$a_{\text{UQFF}}(t) = a_{\min} \cosh\left(\frac{t}{t_{\text{Pl}}}\right) \cdot \left(1 + f_{\text{TRZ}} \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}}\right)^{1/3}$$

The cubic-root term reflects isotropic volumetric expansion from the buoyancy field.

3.3 Effective Equation of State

$$w_{\text{eff}} = -1 + (1 + f_{\text{TRZ}}) \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}} \kappa [\text{SSq}]$$

With $\kappa = 0.0005 \text{ day}^{-1}$ and $[\text{SSq}] = 0.57$:

$$w_{\text{eff}} = -1 + 1.1 \times 10 \times 0.0005 \times 0.57 \approx -1 + 3.135 \times 10^{-3}$$

This is very close to a cosmological constant ($w = -1$) with a small positive deviation consistent with slow quintessence.

3.4 Density Rate

From the UQFF-modified continuity equation:

$$\dot{\rho} = -3H(1 + w_{\text{eff}}) \rho$$

4. UQFF Number System Connections

Number System	PAPER	Connection in PAPER_658
Vacuum Density Series (VDS)	646	ρ_{UA} , ρ_{SCm} define the elevation factor $\times 11$
Dipole Vortex Primes (DVP)	647	Not directly invoked; implicit via μ_{j} in companion PAPER_659
Buoyancy Harmonics (BH Series)	648	Buoyancy pressure elevates bounce from PBH interior outward

5. Numerical Results

Quantity	Standard LQC	UQFF LQC
ρ_{Planck}	$5.16 \times 10^{96} \text{ kg/m}^3$	—
ρ_{c}	$2.12 \times 10^{96} \text{ kg/m}^3$	—
$\rho_{\text{c,UQFF}}$	—	$2.33 \times 10^{97} \text{ kg/m}^3$
Elevation factor	1	$\times 11$
a_{min}	$1.62 \times 10^{-35} \text{ m}$	$1.69 \times 10^{-35} \text{ m} (\times 1.04)$
t_{Planck}	$5.39 \times 10^{-44} \text{ s}$	—
w_{eff}	−1 (exact)	−0.9969
H^2 at $\rho = 0.9\rho_{\text{c,UQFF}}$	0	positive (bounce prevented)

The UQFF elevation means a black hole must compress matter to $11\times$ the standard LQC critical density before the bounce occurs — equivalently, PBHs with masses near the bounce mass survive longer by a factor of ~ 11 .

6. Vacuum Density Series — Derivation

The three Vacuum Density Series terms identified in PAPER_646:

n	Term	Value	Physical Role
1	ρ_{UA}	$7.09 \times 10^{-36} \text{ J/m}^3$	Universal Aether vacuum energy density
2	ρ_{SCm}	$7.09 \times 10^{-37} \text{ J/m}^3$	Superconductive Medium density

n	Term	Value	Physical Role
Ratio	$\rho_{\text{UA}}/\rho_{\text{SCm}}$	10	Bounce elevation factor in LQC

In PAPER_658 these appear as the multiplier on ρ_c and as structural constants in w_{eff} .

7. Simulation Protocol

A simple Euler integrator is implemented in BlackHoleBounceUQFF.cpp:

1. Initialise: $a = a_{\text{UQFF}}(0)$, $\rho = \rho_c \cdot \text{UQFF} \times f_{\text{initial}}$, $dt = t_{\text{Pl}}$
2. Compute H^2 from LQC Friedmann with UQFF ρ_c
3. Update: $\dot{a} = H \cdot a$; $\Delta\rho = -3H(1+w_{\text{eff}})\rho \cdot dt$
4. Output to `lqc_bounce_sim.csv`: t , a , ρ , H^2 , w_{eff}

8. Discussion

The UQFF LQC model makes testable predictions: - **PBH lifetime extension:** PBHs of mass $\sim 10^{15}$ g (currently evaporating) live $\times 11$ longer under UQFF, shifting the peak in the PBH dark matter window. - **CMB imprint:** The elevated bounce scale generates primordial gravitational waves at UQFF-shifted frequencies that may differ from standard LQC predictions by up to $11\times$ in peak amplitude. - **Cosmological constant:** $w_{\text{eff}} \approx -0.9969$ distinguishes UQFF from Λ CDM at the 0.3% level — potentially measurable by Euclid or DESI.

9. Conclusion

The UQFF LQC model provides a physically motivated extension of Loop Quantum Cosmology. By incorporating the Vacuum Density Series constants, the critical bounce density is elevated by a factor of ≈ 11 , with downstream consequences for PBH physics, the equation of state, and primordial gravitational wave signatures. The complete C++ implementation and Python calculator (CP4 #242) enable further numerical exploration.

References

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